

Desert Ten

A little 10-watt solid-state practice combo with an 8-inch open-back speaker, from clean to raspy fuzz.



Overview

Desert Ten is a small 10-watt solid-state practice amp, in the spirit of the little combo the desert-rock records were made with. It has an op-amp preamp that goes from clean to a raspy rail clip, a SATURATION switch that brings in a diode clipper with a midrange push for the gnarly, fuzzy tone, three tone controls, and a 10-watt power amp that stays clean at first and then clips hard.

After the amp comes the 8-inch open-back speaker, with a thin low end and a midrange honk, and a mic you can move from on axis to off axis. Switch the speaker off for a line-out sound to feed your own cabinet.

Like a real amp it runs in mono: the left and right inputs are summed, and both outputs carry the amp. The amp runs at twice the sample rate to keep the clipping clean. A noise gate sits before the amp and a tuner is built in. With SATURATION on, the clipping diodes' balance wanders very slightly over time, as on real hardware.

Reach for it for

- Raspy, fuzzy guitar and bass tones (SATURATION on)
- Small-amp crunch that sits in a dense mix
- Lo-fi, boxy guitar and keys through a little 8-inch speaker

- Driving a DI'd guitar with the gate and tuner on the same panel
- Line-out amp tones into your own cabinet (CAB off)

Controls

GAIN

PRE (cream cap)	The preamp gain, 0 to 10 (5 by default): from about -6 to +30 dB into the op-amp. Low settings are clean; higher settings push the op-amp into its rails for a raspy clip. It also sets how hard the SATURATION diodes are driven.
SATURATION	Switches in the diode clipper with its midrange push: the signal is lifted around 900 Hz, trimmed below 220 Hz and above 5 kHz, and clipped by a pair of diodes. This is the gnarly, fuzzy tone. Off by default.
POST (blue cap)	The volume into the 10-watt power amp, 0 to 10 (4 by default). Low settings stay clean; turned up, the power amp clips hard, the way small solid-state amps do.

EQUALIZATION

LOW	A low shelf at 120 Hz, 0 to 10; 5 is flat, and each step is about 2.6 dB.
MID	A broad peak at 700 Hz, 0 to 10. 5 leaves a small built-in mid lift (about 1.5 dB); turn it down to scoop the mids, up to push them.
HIGH	A high shelf at 2.8 kHz, 0 to 10; 5 is flat.

The lower strip

GATE	A noise gate before the amp, from -80 to -20 dB. Fully down (-80) reads OFF. Raise it until the hiss and hum between notes disappear. It opens fast, holds for 50 ms and closes smoothly.
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TUNER	Switches the tuner on. While it is on, the output is muted so you can tune in silence. The window shows the note name with its octave and a dot that moves left (flat) or right (sharp); the note turns green when you are within a few cents. It reads PLAY A NOTE when it hears nothing.
CAB	The 8-inch open-back speaker, on by default. Off gives the amp's line out, with no speaker colour.
MIC	Where the mic sits on the speaker, 0 (on axis) to 10 (off axis); 3 by default. On axis is brighter with more cone bite; moving off axis rounds off the top end and darkens the sound.
OUTPUT	The final level, -24 to +12 dB.
LEVEL	Ten LEDs showing the output level, up to 0 dBFS.

The header

Theme picker	Changes the panel's colours: the control panel, the pinstripe and the pilot lamp follow the theme. Graphite keeps the aluminium panel with an orange stripe and lamp.
HW	Switches between the hardware-style faceplate and the flat look.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

There are no factory presets, and no input or output fold-out on this panel. The output passes through the GFX soft clipper, which catches peaks at -0.3 dBFS.

Tips

- For the classic fuzzy desert tone, switch SATURATION on, set PRE around 6 - 8 and keep POST moderate; then shape it with MID.
- For a cleaner small-amp crunch, leave SATURATION off and turn POST up: the power amp clips harder than the preamp.
- If the sound is too boxy, move MIC towards 0 and bring MID down a little.
- Set the GATE just above the noise floor with SATURATION on - high gain makes hiss between notes much louder.

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